

Computer Networks

Lab Task 2: Networking Hardware Components/Devices

Networking Components:

The fundamental units that enable device connections, communication, resource sharing and overall make up a computer network are called networking components. Components vary according to network set up. There exist both software and hardware components in a computer network. This document lists the hardware networking components.

The components covered in this document are:

1. Access Point
2. Modems
3. Firewalls
4. Repeater
5. Hub
6. Bridge
7. Switch
8. Routers
9. Gateway
10. Brouter
11. NIC Card

The role of network devices:

1. Data Transmission and reception among devices in a network
2. Network devices increase network speed and control data flow
3. Safeguard network from threats and access control
4. Increase network coverage area and address signal issues

Network hardware components:

The actual parts of a computer network or electrical system that cooperate to carry out different tasks are termed as hardware components.

1. Serves and Clients:
 - a. Clients: gadgets that request other computers (servers) for data or services. They utilize and get access to the network resources.
 - b. Servers offer services to clients, they are strong computers that store information and operate programs that can be accessed by clients. They are responsible for managing network resources.
2. Transmission media:
 - a. The channels that allow data transfer between devices in a network are called transmission media.
 - b. Wired Media: The majority of contemporary Ethernet networks use twisted-pair connections. The physical ethernet LANs often use unshielded twisted pair cable.

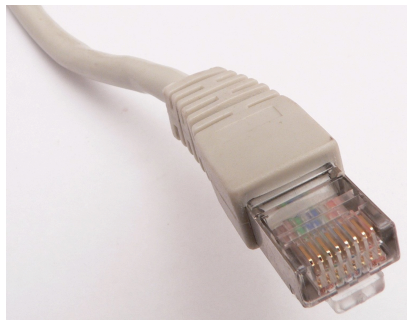
- i. Coaxial cable: Mostly used for audio/video communications, cable television and older network kinds.



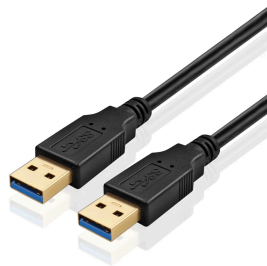
- ii. Fiber-optic cable: They use light pulses to send data, providing fast speeds over large distances.



- iii. Ethernet: The most popular LAN technology



- iv. USB cable: Mainly used to connect PCs and smartphones



c. Wireless Media:

- i. Bluetooth and WiFi radio waves
- ii. Satellite communications uses microwaves
- iii. TV remote controls and other short range communications use IR technology
- iv. Cellular communications

Access Point

It is a device that enables wireless devices like laptops and smartphones to join a wired network. It enables wireless devices to connect to the internet or other networked devices via WiFi network. They are used to provide WiFi in places without it or to increase the network coverage.

Pros:

1. Provides wireless connectivity
2. Extends network coverage

Cons:

1. Speed reduces with many users
2. Requires proper security setup



Modems

A modem is used to transform digital signals into analog signals of various frequencies and send them to a modem at the receiving end. An analog signal can also be converted back to digital signal using modems. Types of Modems: DSL Modem, Wireless Modem

Pros:

1. Enables internet access
2. Converts signals efficiently

Cons:

1. Speed depends on ISP (Internet Service provider)
2. Limited functionality alone



Firewalls

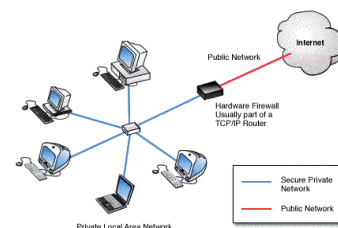
A firewall is a type of network security tool that regulates data traffic between computer networks and the internet. It's a barrier allowing only trusted data to enter the network and prevents unwanted access. Firewalls filter traffic and defend the network against hackers, malware and other online threats.

Pros:

1. Protects Network from attacks
2. Controls unauthorized access

Cons:

1. Needs configuration
2. may block valid traffic



Repeaters

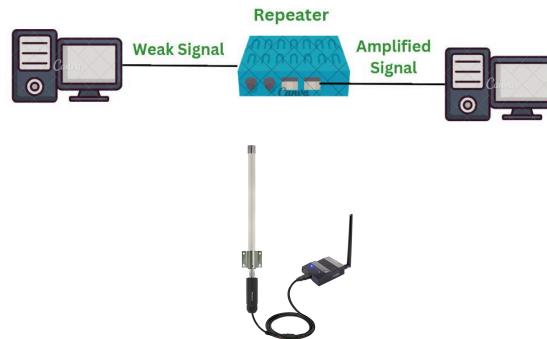
Repeaters are used to regenerate the signal over the same network before it gets too weak or corrupted to broadcast for an extended period. They duplicate the signal bit by bit and regenerate it.

Pros:

1. Extends network distance
2. Improves signal strength

Cons:

1. Cannot filter data
2. May amplify data



Hub

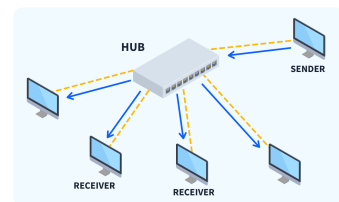
A hub is a repeater with multiple ports. A hub, like the connector in a star topology, links many stations and joins wires from various branches. All hosts connected by hubs maintain a single collision domain. Additionally, they lack intelligence to choose the optimal path from data packets which results in waste and inefficiency. Types of Hubs: Active hub, Passive hub, Intelligent hub.

Pros:

1. Simple and low cost
2. Easy to install

Cons:

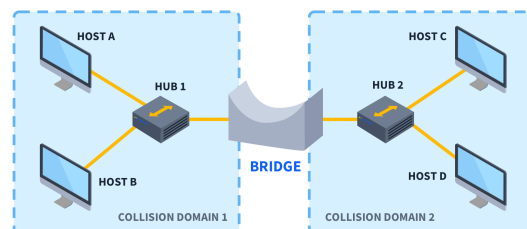
1. High collisions
2. Low security



Bridge

A bridge is a repeater that filters by reading source and destination MAC addresses. Connecting 2 LANs that have the same protocol is another use case of it. Each port is connected to a distinct network segment, and it usually connects several segments. For the purpose of connecting and managing various network segments, a bridge can include more than two ports. Because they serve comparable purposes, contemporary multi-port bridges.

Types of Bridges: Transparent Bridges, Source Routing Bridges



Pros:

1. Reduces network traffic
2. Improves performance

Cons:

1. Limited scalability
2. Slower than switches

Switch

A switch connects multiple devices within a network and forwards data only to the intended device, improving speed and security compared to a hub.

Types of Switches: Unmanaged switches, Managed switches, Smart switches, Gigabit switches, Rack-mounted switches, Desktop switches.

Pros:

1. High speed data transfer
2. Reduces collisions

Cons:

1. Costlier than hubs
2. Managed switches need configuration



Router

A router is like a switch, it routes data packets by IP address. A router is primarily a Network Layer device. Your router's dynamic routing table routes data packets between LANs and Wide Area Networks(WANs). Router-connected hosts have independent broadcast domains.

Pros:

1. Connects different networks
2. Supports internet sharing

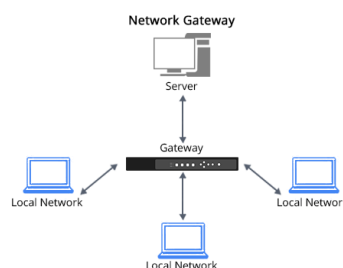
Cons:

1. Expensive
2. Complex setup



Gateway

A gateway is a passageway that connects two networks that may operate using various networking architectures. As messenger agents, they collect information from one system, decipher it, and then send it to another. Gateways can function at any network layer and are also known as protocol converters. Compared to switches or routers, gateways are typically more complicated.



Pros:

1. Connects different network architectures
2. Protocol conversion

Cons:

1. Complex design
2. High processing load

Router

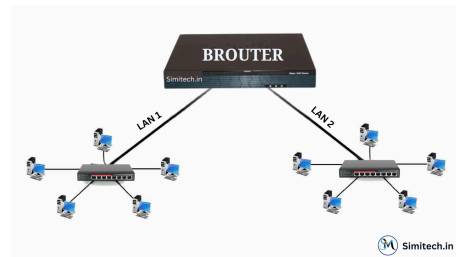
Bridging routers combine bridge and router functionalities. It works at the network or data connection layer. It can function as a bridge to filter traffic from local area networks and as a router to transport packets across networks.

Pros:

1. Combines routing and bridging
2. Flexible operation

Cons:

1. Rarely used
2. Complex configuration



NIC Card

A network adapter called a network interface card, or NIC, is used to link a computer to a network. In order to create a LAN, it is installed on the computer. It features a connector to attach the cable to it, as well as a unique ID printed on the chip. The cable serves as a conduit between the modem or router and the PC.

Pros:

1. Essential for network access
2. Provides unique MAC address

Cons:

1. Hardware failure stops connectivity
2. Requires drivers



WiFi Adaptor

It allows a computer to connect to a wireless network. It can be internal or external (USB-based).

Pros:

1. Enables wireless internet access
2. Portable and easy to use, no cables necessary

Cons:

1. Slower than wired connection
2. Affected by distance and interference



RJ45 Connector

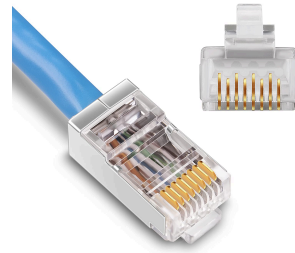
RJ45 is a standard connector used to connect Ethernet cables to networking devices like computers, switches, and routers.

Pros:

1. Widely used and standardized
2. Easy to install and cost-effective

Cons:

1. Limited to short distances
2. can be damaged easily



CAT5 Cable

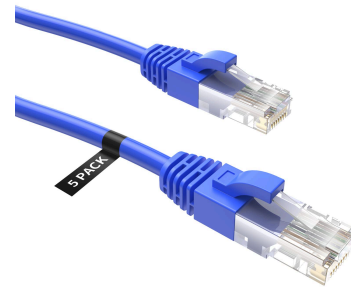
CAT5 cables are twisted-pair Ethernet cables used for basic networking and data transmission.

Pros:

1. Low cost
2. easy to install

Cons:

1. Lower speed (up to 100 Mbps)
2. not suitable for modern high-speed networks



CAT6 Cable

CAT6 cables support higher data speeds and reduced interference compared to CAT5.

Pros:

1. Supports high-speed network (1Gbps+)
2. Better noise resistance

Cons:

1. More expensive than CAT5
2. thick and less flexible



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